

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

- Classwork 3
  - Due this **Friday 9/27 @5 pm**
  - Typo: Problem 2.3: “using your work from parts 2.1 and 2.2”
- Homework 1
  - Due Friday 10/4 @5 pm

This Lecture

- Utility Theory
  - Utility Function
  - Expected Utility and Maximum Expected Utility
- Decision Networks
  - Bayesian Network & Utility Theory
- Q&A for Classwork 3/Homework 1

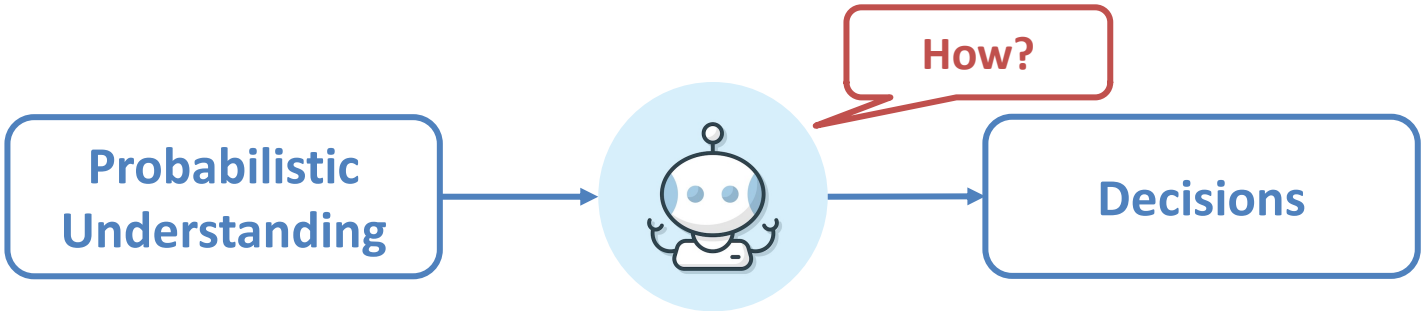
Last Lecture

- Inference goal of BN
  - $BN.ask(Q, e)$=$P(Q|e)$
- Compute $P(Q|e)$ given BN?
  - **Exact** Inference
    - Queries can be efficiently computed in O(1), O(|V|), O($|V|^2$)
  - **Enumeration** Inference
    - Computable queries that could not be efficiently computed
    - Sums-of-products of conditional probabilities from the CPTs

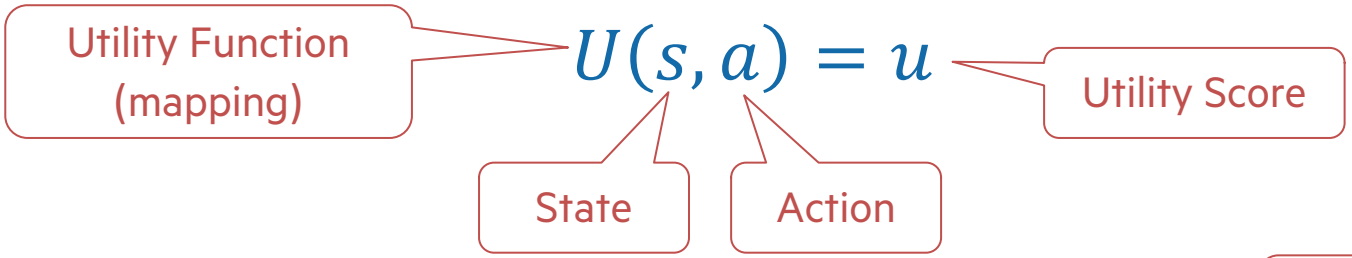
Probabilities -> Decision

- If you are a logical person, which lottery will you choose?





Utility Function



- Example: $30k lottery, ticket price = $1

Is something missing here?

U(s,a)		Action a	
		a = 0 (did not buy)	a = 1 (bought)
State s	s = 0 (did not win)	0	-1
	s = 1 (won)	0	30k-1

Expected Utility

- "weighted avg utility score" = Utility Function + Probability
- **Expected Utility** (EU) of some action a given any evidence e is the **average utility value** of the possible outcome states $s \in S$ **weighted** by the probability that each outcome occurs:

$$EU(a|e) = \sum_{s \in S} p(s|a,e) \cdot U(s,a)$$

likelihood

utility

Expected Utility

$$EU(a|e) = \sum_{s \in S} p(s|a,e) \cdot U(s,a)$$

- Example
  - $30k lottery, ticket price = $1, **% of winning = 0.0000001**
  - What are $EU(a=1)$ and $EU(a=0)$?
  - $EU(a=1) = p(s=0|a=1) \cdot u(s=0, a=1) + p(s=1|a=1) \cdot u(s=1, a=1)$
    $= 0.9999999 * (-1) + 0.0000001 * 29{,}999$
    $= -0.997$
  - $EU(a=0) = 0$

e={}

How could EU help with decision making?

Maximum Expected Utility

- An agent faced with $a \in A$ actions with uncertain outcomes (and any observed evidence e) shall choose the action $a*$ that **maximizes EU**:

float

action

$$MEU(e) = \max_{a \in A} EU(a|e)$$

$$a^* = argmax_{a \in A} EU(a|e)$$

best action

choose an action a that maximizes EU

Maximum Expected Utility

- Example
  - $30k lottery, ticket price = $1, **% of winning = 0.0000001**
  - EU(a=1) = -0.997 and EU(a=0) =0
  - What is MEU and a*?
  - $MEU(e = \emptyset) = \max_a EU(a)$

$$= \max(EU(a = 1), EU(a = 0))$$
$$= \max(-0.997, 0) = 0$$

  - $a^* = argmax_{a \in A} EU(a) = 0$

action a*=0 that leads to max EU, i.e., EU(a) = 0

Next …

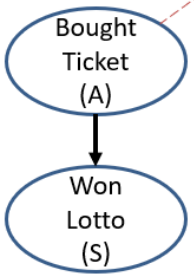
DECISION NETWORKS

Decision Networks

- Bayesian Networks + Utility Theory
- Determine which variables should be scored and which variables constitute actions.
- How? – Adding two additional nodes in BN
  - **Chance** Nodes (◯): traditional BN variables with CPTs
  - **Decision** Nodes (□): choices that the agent has to make
  - **Utility** Nodes (◇): utility functions

Decision Networks

As Bayesian Network (BN)

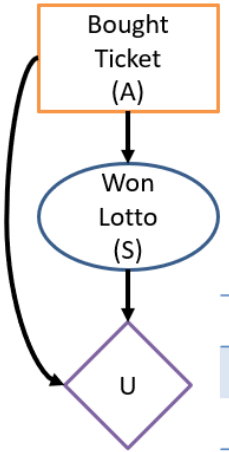


What would $P(A)$ look like?

Full Joint: $P(A,S)$

$P(S|a,e)$ query: $P(S|a) = \frac{P(S,a)}{P(a)}$

As Decision Network (DN)



Full Joint: $P(S|do(A))$

$P(S|a,e)$ query: $P(S|do(a))$

A	S	$P(S \mid do(A))$
0	0	1.0000000
1	0	0.9999999

$U(pa) = U(A,S)$ = (same as example above)

Decision Nodes

- Have **no parent**
- Have **no CPT** (assumed "given")
- Do **not** need to be **normalized**
- A.k.a. "intervention"

Utility Nodes

- **Parents** of utility nodes (pa) are the arguments to the utility function.
- **Parents** of utility nodes can be **chance** nodes OR **decision** nodes.

Expected Utility for DN

$$EU(a|e) = \sum_{s \in S} p(s|a,e) \cdot U(s,a)$$

- How to compute EU for DN?

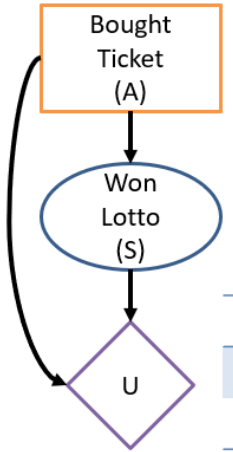
a: full assignments to decision node

$$EU(a|e) = \sum_{s \in S} p(s|do(a),e) \cdot U(pa)$$

e: **observations** about the chance nodes

s: "state" or chance node **parents** of utility nodes

As Decision Network (DN)



Full Joint: $P(S|do(A))$

$P(S|a,e)$ query: $P(S|do(a))$

A	S	$P(S \mid do(A))$
0	0	1.0000000
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$U(pa) = U(A,S) =$ (same as example above)